

Yongjun Rong

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Summary

A Principal AI Architect with deep, hands-on expertise in Generative AI, Agentic Systems, and Multi-modal AI. I specialize in architecting and delivering advanced AI products from concept to deployment, with a proven track record in building multi-agent frameworks, RAG systems, and scalable AI solutions on cloud platforms. My work integrates diverse data types into unified AI-driven applications, leveraging cutting-edge tools like LangChain, OpenAI, and PyTorch/TensorFlow to solve complex problems in healthcare, finance, and technology. I am a strategic leader who remains deeply technical, focused on driving real-world AI innovation and mentoring teams to excel in modern AI development.

Tags: Principal Architect, GenAI, Agentic AI

Skills

Generative AI & Multimodal Systems: Extensive expertise in generative AI, including LLMs, RAG, and multi-agent frameworks (Autogen), with a strong ability to design systems that process and unify text, images, and audio for sophisticated AI applications.

AI Product Architecture & Deployment: Proven ability to architect and deliver complex, market-ready AI-driven products. Hands-on experience in launching GenAI products, from initial concept and agentic system design to full-scale deployment and optimization on cloud platforms.

AI Software Development: Strong proficiency in Python and AI frameworks like LangChain, OpenAI API, PyTorch, and TensorFlow. Expert in applying software design patterns, clean architecture, and SOLID principles to build scalable, maintainable AI systems.

Cloud & Scalable AI Infrastructure: Expertise in cloud platforms (AWS, GCP, Azure) with a focus on building distributed, high-performance AI architectures. Deep experience with serverless computing (Lambda, EventBridge), containerization (Docker, Kubernetes), and Infrastructure as Code (Terraform) for scalable AI solutions.

Technical Leadership & Strategy: A visionary leader who drives AI strategy, fosters innovation, and mentors engineering teams. Experienced in establishing best practices, leading architecture reviews, and making strategic technology decisions for AI investments.

Technologies

AI Frameworks: LangChain, OpenAI API, PyTorch, TensorFlow, RAG, Multi-agent (Autogen)

Programming Languages: Python, Java, Terraform, Bash

Cloud Platforms: AWS (Lambda, EventBridge, SQS, SNS, EKS), GCP, Azure

Work Experience

Director of Cloud Architecture

Feb. 2025 – Current

Biophy, Cherry Hill, NJ (WFH), USA

As a hands-on director, I architected and implemented a comprehensive AI-powered RAG platform from the ground up. I led the development of the core data ingestion, agentic scraping system, and a

custom alerting framework, applying advanced AI principles and clean architecture to deliver a scalable, production-ready GenAI product.

Technologies: Python/Django/Pydantic, openAI API/LangChain, Agentic AI, RAG, LLMs, AWS/Lambda/SQS, Infrastructure as Code (Terraform/Terragrunt), SOLID Principles, Clean Architecture

Project: AI-Powered Data Ingestion and Scraping System for RAG

- Architected a modular data ingestion pipeline supporting multiple data sources and formats for multimodal data processing.
- Implemented intelligent, agentic web scraping capabilities with rate limiting and error handling.
- Designed data transformation layers following Clean Architecture principles to prepare data for LLM consumption.
- Built robust error handling and retry mechanisms for data processing failures to ensure system resilience.
- *Leadership:* Drove the adoption of Clean Architecture and SOLID principles across the data engineering and AI teams. Established data quality standards and monitoring practices for AI system reliability. Mentored developers on modern AI software design patterns and best practices.

Senior Cloud Engineer II (Associate)

Dec. 2021 – Feb. 2025

McKinsey & Company, Cherry Hill, NJ (WFH), USA

Led the architecture and development of advanced genAI systems with RAG and multi-agent frameworks to accelerate complex technical migrations and data pipeline development. Scaled RAG vector-stores for real-time chatbot copilots and successfully architected the migration of complex systems to modern AI-ready, serverless architectures.

Technologies: genAI/openAI/Langchain/RAG/Multi-agents/Autogen, VectorStore/ChromaDB/Milvus, Python/Flask/FastAPI/Pydantic/Jupyter, Agents/Multi-agents, Event Driven Architecture (EDA), AWS/EventBridge/Lambda/Step Functions, Docker/K8s/EKS

Projects:

- **Using genAI (openAI) with multi-agents to accelerate the migration of mainframe to modern technologies**
 - Core contributor to genAI system to migrate mainframe technologies to modern cloud-native stacks.
 - Contributed to developing multi-agent genAI system (e.g., Autogen) to automate the development/testing/deployment process.
 - Contributed to developing RAG system to accelerate the migration process by using ChromaDB/Milvus to store the dependency context for agents.
 - *Leadership:* Drove the integration of multi-agent genAI systems to automate legacy migration, reducing manual effort and error rates. Collaborated across teams to ensure seamless modernization from mainframe to AI-native stacks. Championed the use of RAG for context-aware agentic systems, setting a new benchmark for AI-driven modernization projects.
- **Using genAI (openAI) with RAG to accelerate the development of data pipelines**
 - Prompts tuning and engineering to generate data pipeline function code and unit test code.
 - Designed and implemented chunking system to split large inputs for LLM context window management.
 - Designed and implemented parallel calls to openAI API by using asyncio to improve AI workflow performance.

Education

Master of Computer Science

Texas Tech University, Lubbock, TX, USA

GPA: 4.0/4.0

Sept. 2002 – Jun. 2006

Master of Engineering

Shanghai JiaoTong University, Shanghai, China

Major: Hydraulic Fluid Control in Mechanical Engineering

Sept. 1993 – Feb. 1996

Languages

English: Fluent

Chinese: Native

Hobbies

Playing with cutting-edge technologies: genAI/Autogen/Multiagents, Docker, Kafka, Kubernetes

IoT or Robot programming: Programming against iRobot create SDK and MIPOSaur SDK, exploring agentic behaviors in robotics.